```
"use strict";

const $ = (id) => document.getElementById(id);

const scene = $("scene");
const runner = $("runner");
const coinTrack = $("coinTrack");

let running = false;
let paused = false;
let lane = 0;
let coins = 0;
let distance = 0;
let speed = 1;

let raf = null;
let lastTime = 0;
let coinTimer = 0;
let obstacleTimer = 0;
let activeCoins = [];
let obstacleY = -100;
let obstacleLane = 0;

const lanePositions = [29, 50, 71];

function setLane() {
  runner.style.left = lanePositions[lane + 1] + "%";
}

function showToast(message) {
  $("toast").textContent = message;
  $("toast").classList.remove("hidden");

  setTimeout(() => {
    $("toast").classList.add("hidden");
  }, 1400);
}

function startGame() {
  cancelAnimationFrame(raf);

  running = true;
  paused = false;
  lane = 0;
```

```
  coins = 0;
  distance = 0;
  speed = 1;
  coinTimer = 0;
  obstacleTimer = 0;
  obstacleY = -100;
  activeCoins = [];

  $("coins").textContent = "0";
  $("distance").textContent = "0";

  coinTrack.innerHTML = "";
  $("obstacle").style.display = "none";

  $("overlay").classList.add("hidden");
  $("gameover").classList.add("hidden");

  $("pauseBtn").textContent = "⏸";

  runner.className = "runner";

  setLane();

  lastTime = performance.now();

  raf = requestAnimationFrame(loop);

  showToast("Collect coins and dodge the rocks!");
}

function loop(now) {
  if (!running || paused) return;

  const dt = Math.min((now - lastTime) / 16.67, 2);

  lastTime = now;

  distance += 0.16 * speed * dt;

  speed = 1 + distance / 850;

  $("distance").textContent = Math.floor(distance);

  coinTimer += dt;
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obstacleTimer += dt;

if (coinTimer > 32) {
  coinTimer = 0;
  spawnCoin();
}

if (obstacleTimer > 105 / speed) {
  obstacleTimer = 0;
  spawnObstacle();
}

activeCoins = activeCoins.filter((coin) => {
  coin.y += 2.1 * speed * dt;

  coin.el.style.top = coin.y + "%";

  if (
    coin.y > 75 &&
    coin.y < 94 &&
    coin.lane === lane
  ) {
    coins++;

    $("coins").textContent = coins;

    coin.el.remove();

    return false;
  }

  if (coin.y > 110) {
    coin.el.remove();
    return false;
  }

  return true;
});

if (obstacleY >= 0) {
  obstacleY += 1.45 * speed * dt;

  $("obstacle").style.top = obstacleY + "%";
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    $("obstacle").style.left =
      lanePositions[obstacleLane + 1] + "%";

    const hitObstacle =
      obstacleY > 76 &&
      obstacleY < 92 &&
      obstacleLane === lane &&
      !runner.classList.contains("jump") &&
      !runner.classList.contains("slide");

    if (hitObstacle) {
      endGame();
      return;
    }

    if (obstacleY > 110) {
      obstacleY = -100;

      $("obstacle").style.display = "none";
    }
  }

  raf = requestAnimationFrame(loop);
}

function spawnCoin() {
  const randomLane = Math.floor(Math.random() * 3) - 1;

  const coin = document.createElement("span");

  coin.className = "coin";
  coin.textContent = "🪙";

  coin.style.left =
    lanePositions[randomLane + 1] + "%";

  coin.style.top = "0%";

  coinTrack.appendChild(coin);

  activeCoins.push({
    el: coin,
    y: 0,
    lane: randomLane
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  });
}

function spawnObstacle() {
  if (obstacleY >= 0) return;

  obstacleLane = Math.floor(Math.random() * 3) - 1;

  obstacleY = 30;

  $("obstacle").style.display = "block";

  $("obstacle").textContent =
    Math.random() > 0.5 ? "🪨" : "🪵";

  $("obstacle").style.left =
    lanePositions[obstacleLane + 1] + "%";
}

function move(direction) {
  if (!running || paused) return;

  lane = Math.max(
    -1,
    Math.min(1, lane + direction)
  );

  setLane();
}

function jump() {
  if (!running || paused) return;

  runner.classList.remove("slide");
  runner.classList.add("jump");

  setTimeout(() => {
    runner.classList.remove("jump");
  }, 650);
}

function slide() {
  if (!running || paused) return;
```

```
  runner.classList.remove("jump");
  runner.classList.add("slide");

  setTimeout(() => {
    runner.classList.remove("slide");
  }, 650);
}

function endGame() {
  running = false;

  cancelAnimationFrame(raf);

  $("finalDistance").textContent = Math.floor(distance);
  $("finalCoins").textContent = coins;

  $("gameover").classList.remove("hidden");
}

$("startBtn").addEventListener("click", startGame);

$("restartBtn").addEventListener("click", startGame);

$("pauseBtn").addEventListener("click", () => {
  if (!running) {
    showToast("Press START RUN to begin");
    return;
  }

  paused = !paused;

  $("pauseBtn").textContent = paused ? "▶" : "⏸";

  if (!paused) {
    lastTime = performance.now();

    raf = requestAnimationFrame(loop);
  }
});

document.addEventListener("keydown", (event) => {
  if (
    ["ArrowLeft", "ArrowRight", "ArrowUp", "ArrowDown", " "]
      .includes(event.key)
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  ) {
    event.preventDefault();
  }

  if (event.key === "ArrowLeft" || event.key.toLowerCase() === "a") {
    move(-1);
  }

  if (event.key === "ArrowRight" || event.key.toLowerCase() === "d") {
    move(1);
  }

  if (event.key === "ArrowUp" || event.key === " ") {
    jump();
  }

  if (event.key === "ArrowDown") {
    slide();
  }
});

let touchStart = null;

scene.addEventListener("touchstart", (event) => {
  const touch = event.changedTouches[0];

  touchStart = {
    x: touch.clientX,
    y: touch.clientY
  };
}, { passive: true });

scene.addEventListener("touchend", (event) => {
  if (!touchStart) return;

  const touch = event.changedTouches[0];

  const dx = touch.clientX - touchStart.x;
  const dy = touch.clientY - touchStart.y;

  if (Math.max(Math.abs(dx), Math.abs(dy)) < 25) {
    touchStart = null;
    return;
  }
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  if (Math.abs(dx) > Math.abs(dy)) {
    move(dx > 0 ? 1 : -1);
  } else if (dy < 0) {
    jump();
  } else {
    slide();
  }

  touchStart = null;
}, { passive: true });
```